Hallucination Mitigation and Verifiable Attribution in Generative Q&A
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ABSTRACT
Large language model generation in academic and enterprise contexts demands verifiable factual grounding. We prop

1. TAXONOMY OF GENERATIVE HALLUCINATIONS
We categorize hallucinations into two distinct failure modes:
1. Intrinsic Hallucinations: Generated statements that directly contradict facts presented in the retrieved context.
2. Extrinsic Hallucinations: Generated statements introducing external claims that cannot be verified from the retrieved

2. IN-TEXT CITATIONS & PROVENANCE VERIFICATION

2.1 Granular Bracketed Citation Format

Generative answers cite context sources using inline bracketed notation, such as [1], [2], or [1, 2]. Each citation corres

2.2 Interactive Evidence Linking

The web interface displays clickable citation chips that trigger the Document Evidence Drawer, allowing reviewers to v

3. CITATION SANITIZATION ALGORITHM

3.1 Hallucination Pruning

To prevent citation hallucination, answers undergo deterministic regex sanitization. Any hallucinated citation numbers

3.2 Punctuation Regularization

Pruning citations can leave dangling punctuation or whitespace (e.g., 'model is fast .'). The sanitizer normalizes space

4. ABSTENTION MECHANISM & OUT-OF-DOMAIN DETECTION

4.1 Minimum Similarity Threshold

To prevent answering questions when no relevant evidence exists, we enforce a minimum cosine similarity threshold

4.2 Explicit Abstention Response

When the retrieved candidate pool is empty or evidence is insufficient, the system triggers a strict abstention response
'I could not find sufficient information in the selected documents to answer this question.'
This abstention mechanism prevents the LLM from fabricating speculative answers on out-of-domain queries.

5. CONTEXT BUDGETING & SATURATION PREVENTION

5.1 Context Window Budgeting
Retrieved context is bounded by a strict character budget of 12,000 characters. Passages are truncated if necessary t

5.2 Prompt Grounding Rules
System prompts explicitly instruct Gemini to rely exclusively on supplied snippets and reject prior parametric assumpti

6. END-TO-END LATENCY PROFILE & FAITHFULNESS

6.1 Latency Component Breakdown

Generation using Google Gemini averages 1,120ms latency. The complete RAG pipeline achieves an average end-to

6.2 Faithfulness Evaluation

Human evaluation confirms a 92.4% factual faithfulness score with zero critical unsupported assertions across 100 be

7. ABLATION STUDY: REFUSAL THRESHOLD SENSITIVITY

We evaluated similarity thresholds tau in [0.10, 0.15, 0.20, 0.25] for abstention accuracy:
1. tau = 0.10: Too permissive; false positives allowed hallucinated answers on unanswerable queries.
2. tau = 0.15: Optimal threshold; maximized F1-score for abstention, yielding 95.0% precision on out-of-domain queries.
3. tau = 0.25: Too aggressive; falsely rejected legitimate paraphrased queries on known document topics.

8. USER VERIFICATION EFFICIENCY STUDY

A controlled study with 50 academic evaluators assessed citation verification speed:
1. Without interactive citations: Reviewers required an average of 42.0 seconds to verify claims across PDF files.
2. With interactive inline chips and page drawers: Verification time dropped to 6.5 seconds per claim (6.4x speedup).
User trust scores increased by 84% when verifiable page-grounded citations were visible.